

Displacement Controlled Actuator Cyclic Loading Test Setup

Objective:

Perform horizontal cyclic load test, which simulates loads of earthquakes upon columns and piles made of RC or steel framework.

Complete Specifications:

Double acting Top Loading Cylinder (100 KN Static Rating and 100 mm stroke) with adjustable loading, 100 KN Load Cell fitted on hydrostatic bearings, both being driven by a 250 bar/ 2lpm Mini Hydraulic Power Pack with proportional pressure control valve and displacement controlled double acting horizontal actuator (130 KN static and 100 KN Dynamic rating and 300 mm stroke for a max. amp. of ± 150 mm and max. frequency of 50 Hz) fitted with sealless Piston on hydrostatic bearings, accumulator, all being driven by a 325 bar/100 lpm. Main Hydraulic Power Pack and controlled by a Moog Servo Valve along with Controller, PC based Data Acquisition System (min. 16 Ch. I/O) with 'NI' make 'Lab View' software, Printer, LVDT, 100 KN Fatigue Rated Load Cell etc.

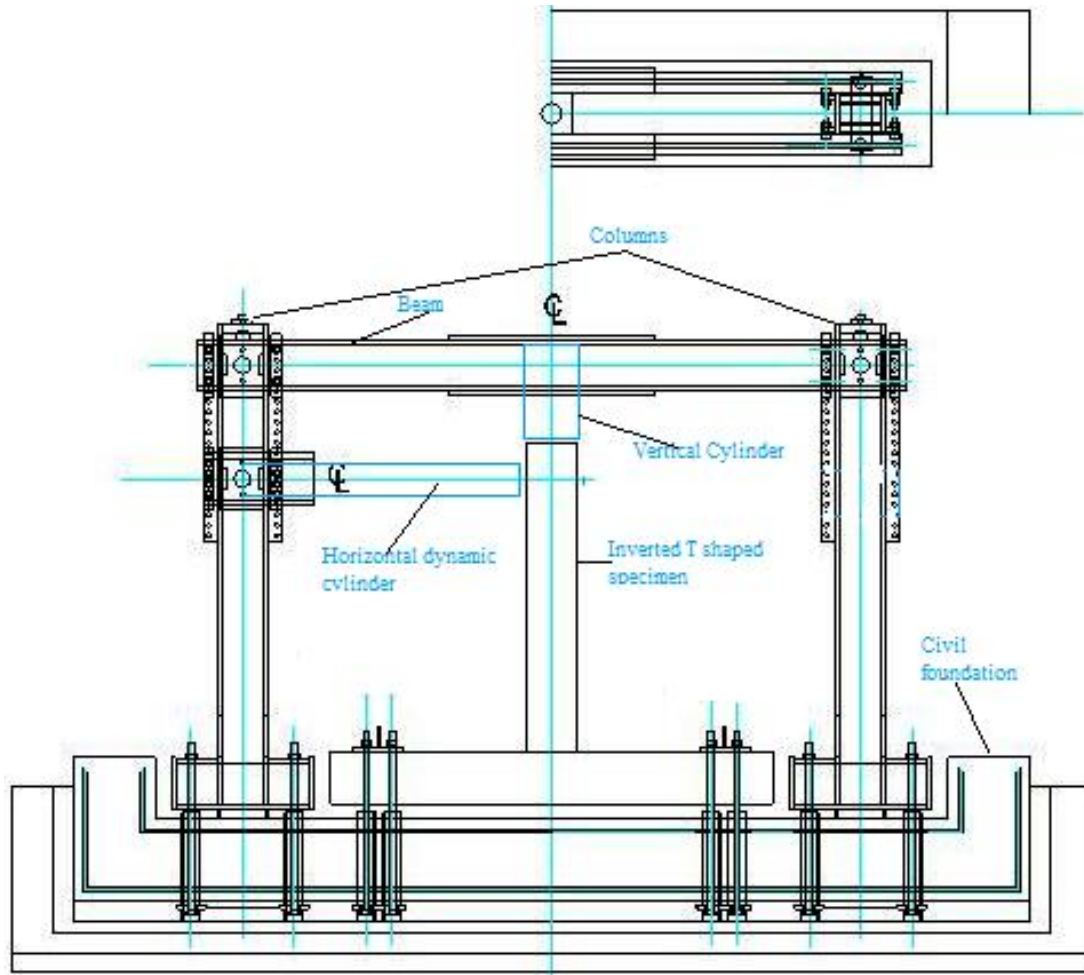


Figure [1]: Cyclic Test Rig general layout



Figure [2]: Mechanical Hardware Setup of Shake Table and Cyclic Loading Test Setup

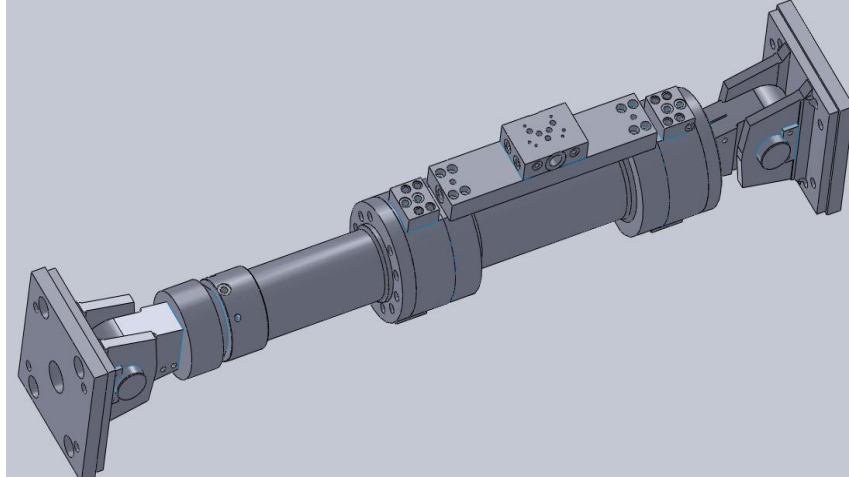


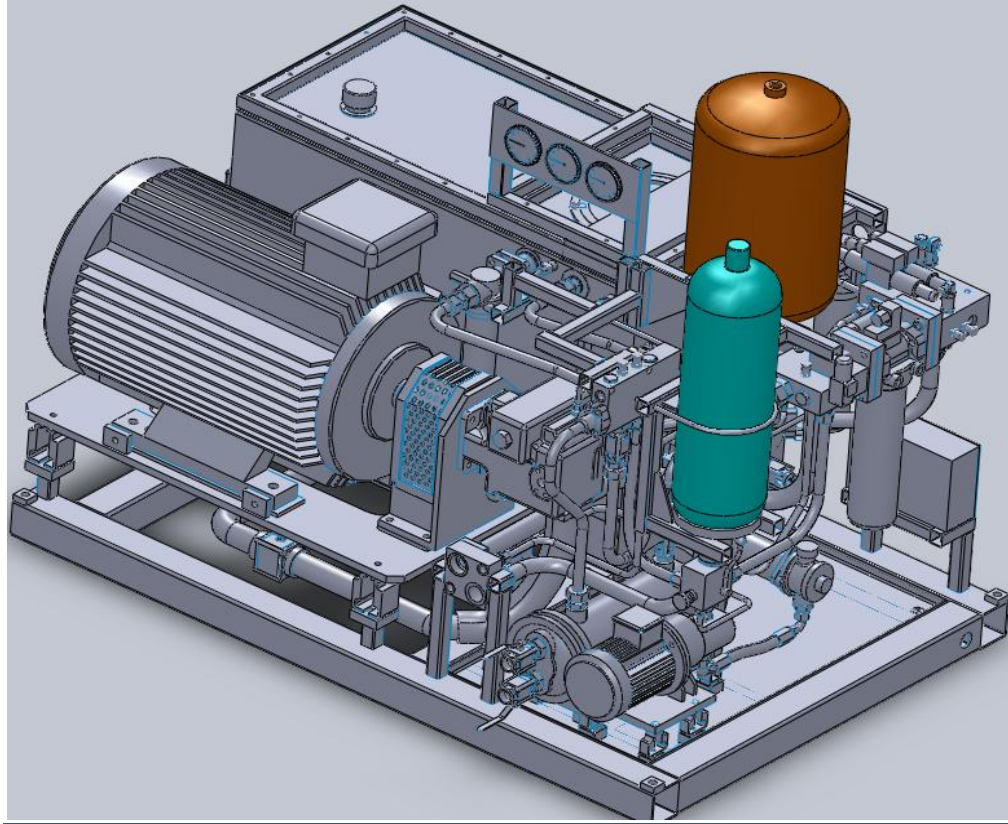
Figure [3]: 3D CAD Model of Dynamic Cylinder and Characteristics Motion of the Dynamic Actuator

Description of hydraulic power pack:

- Electric motor: It provides the power for running the pump.
- Pump: It pumps the oil to a higher pressure from the tank.
- Filter: It filters the oil so that the impurities present in it does not damage the servo valve.
- Storage tank/accumulator: It stores the oil.
- Surge tank: It provides the extra oil required during a sudden rise in pressure and stores the extra oil during sudden drop in pressure.
- Direction control valve: It controls the fluid flow through the circuit.
- Pressure gauges : It measures the pressures in the pipelines.



Fig[3]: Schematic Layout of the Circuit of Main Hydraulic Power Pack



Fig[4]: 3D CAD Model of the Main Hydraulic Power pack

Sensors, Instrumentation and Data Acquisition

For data acquisition (dynamic load, displacement and acceleration) from shake table and cyclic loading test setup, the high precision load cell – HBM Germany, Displacement Transducer / LVDT – HBM Germany / Honeywell Sensotec, and accelerometer - Kistler has been procured. The followings are the photographs of different sensor for shake table and cyclic loading test setup

Dynanic Signal Acquisition Academic Bundle LabView for Windows software PCI-4472 (24 Bit 102.4 ks/s, 8 input, 110 db, dynamic range, sound and vibration toolkit including octave, analysis, FFT, FRE, THD, IMD etc.

Ni PCI 6221 M Series DAQ (16 Analog input 24 Digital I/O 2 Analog Output)
with NI DAQ mx driver software.

SCXI-1540(8 Channel LVDT module)

SCXI-1000 DC (4 slot chassis DC powered)

Chassis powered from any 9 to 16 v DC source or optional battery pack or AC
adaptor.

SCXI-1374 handle kit for PXI-1010 kit includes one handle for side of SCXI
chassis.

SCXI-1520 8 channel universal strain / bridge module kit.

SCXI-1315 Front mounting terminal block

SCXI-1314 Front mounting terminal block (serial terminal)

SCXI-1349 Bracket / Adaptor assembly

SHC68-68 EPM shielded cable, 68-D type to 68 VHDCI offset 0.5 m

SCXI-1361 Rear filter panel cover rear of one slot of SCXI-1000 SCXI-1001 or
SCXI-1000 DC

The National Instruments SCXI-1540 8-channel linear variable differential
transformers (LVDT) module provides the necessary conditioning to measure
signals from transformer-based ratiometric position sensors, including LVDTs,
rotary variable differential transformers (RVDTs), and resolvers.

NI SCXI-1540 - SCXI 8-Channel LVDT Input Signal Conditioning

- 8 channels
- Programmable input range per channel (0.05 to 6 Vrms)
- Programmable 1 or 3 Vrms excitation per channel at 2.5, 3.3, 5, or 10 kHz
- Multichannel synchronization
- 333 kS/s maximum sampling rate (250 Hz output bandwidth)

SCXI Universal Strain Gauge Input Module

- 8 simultaneously sampled analog input channels
- Programmable excitation (0-10 V) per channel
- Programmable gain (1 to 1000) per channel
- Programmable 4-pole Butterworth filter (10 Hz, 100 Hz, 1 kHz, 10 kHz) per channel
- Quarter, half, and full-bridge complete

Discussion

The power is turned on. Then the motor is turned on. The motor drives the pump, which develops the oil pressure. The flow control valve of the delivery channel is kept closed during this, pressure increases until it reaches desired level. Then the flow control valve is opened and oil is supplied to the displacement controlled double acting horizontal actuator via a servo controlled valve and also to the hydrostatic bearings. By suitably opening and closing the servo valve the desired amplitude and frequency of the vibration of the loaded column can be controlled.